D03 · Shop Floor Execution & MES

04 · Shop floor data collection (manual/automated)

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1 WHAT IT IS

Shop floor data collection is the instrumentation layer that feeds the MES — barcode scanning, RFID, IoT sensors, direct PLC integration, and (least preferred) operator manual entry. The discipline is to capture the data the system needs without converting the operator into a data entry clerk; the failure mode is the opposite, and it is catastrophic for both data quality and labor productivity.

2 WHY IT MATTERS

The shop floor data collection layer determines whether the MES is fed truth or fiction. **Every minute an operator spends entering data is a minute subtracted from physical work, and every error in that entry compounds downstream as wrong inventory, wrong traceability, wrong OEE**. The asymmetry is severe — the data entry burden falls on the operator (the lowest-paid worker in the value chain) while the data consumers are higher up the organizational stack. The economic case for automated data capture is rarely questioned **(McKinsey, 2023 — industry analysis)**; what gets questioned is the upfront investment in sensors and integration, which is precisely where the savings live. **The data entry tax is invisible because it is paid in seconds per transaction, but it scales linearly with production volume**.

3 HOW IT IS DONE

1. **Inventory every piece of data the MES asks the operator to enter — and challenge each one.** A typical plant collects 30–60 data fields per work order, of which ~half were specified by someone who left the company three years ago. The first move is the audit: which fields are still used, which can be eliminated, which can be automated.
2. **Automate before instrumenting — eliminate the data point if possible.** The cheapest data to collect is the data you do not need to collect. Many manual entries can be derived from other captured data (start time + cycle time = end time) or made unnecessary by changing the process. Only after this exhaustion does sensor investment make sense.
3. **Use the right capture technology for the data type — barcode, RFID, IoT, PLC, scale, vision.** Material movement is barcode or RFID; machine state is direct PLC; weight is scale integration; quality count is vision. There is no single right technology; there are right technologies per data type, and using the wrong one creates either bad data or bad ergonomics.
4. **Design capture into the operator's existing motion, not as an additional step.** The best data capture is the kind the operator doesn't notice. RFID-tagged kits scanned by a gantry as they pass; PLCs reporting state without prompts; weight scales that capture as parts are placed. If the capture requires an operator to stop, decide, and enter, the design is wrong.
5. **Validate at the source — never accept impossible values downstream.** If the system accepts a defect count of -3 because the operator fat-fingered, the data is corrupted before any analysis runs. Range validation, plausibility checks, and pattern detection at the point of capture prevent garbage from entering the database.
6. **Audit the data quality KPI continuously — not at year-end.** Data quality decays: sensors drift, barcodes wear, operator workarounds emerge. A weekly data quality dashboard (completeness %, timeliness, validation failure rate) keeps the issue visible. Without it, the MES gradually fills with noise that nobody notices until the analysis is wrong.

WORKED EXAMPLE — ILLUSTRATIVE

A pharmaceutical plant audits its shop floor data capture and finds 47 data fields collected per batch — of which 9 are still actively used, 14 are derivable from other captured data, 7 have not been queried in 18 months, and the remaining 17 are mandatory for regulatory reasons. The implementation: 16 fields automated via PLC and scale integration, 7 fields eliminated, 7 replaced with derived calculations, 9 reduced to barcode scan. Operator data entry time per batch drops from **11.4 minutes to 1.6 minutes**; data validation failures down 78% (most caught at the point of capture); regulatory audit findings on data integrity drop from 5 per year to 0 the following audit. Annualized labor savings: **~2,200 hours; cost of sensor implementation $340K; payback under 8 months**.

IN PRACTICE

Operations owns data-capture design, not IT alone

What data we need, captured how, validated where — these are operations design decisions, not tooling decisions. The IT team plugs sensors into MES; the operations team decides which sensors and why.

Audit and eliminate fields before instrumenting them

Most plants instrument data they no longer use. Walk every capture point twice a year with a cross-functional team — operator + supervisor + automation engineer + MES configurator — and kill the dead fields before automating them.

Validate at the source, not downstream

An error caught at scan time costs seconds to fix; the same error caught in inventory reconciliation costs hours **(McKinsey, 2023 — industry analysis)**. The MES should reject implausible entries immediately — wrong material code, out-of-range quantity, mismatched lot — not log them for later.

Match capture technology to data type and motion

Barcode/RFID for material movement, vision for quality, PLC-direct for cycle data, IoT for environmental — and design the capture into the operator's existing motion, not as a separate step. The data-entry station that requires a walk is the data-entry station that lies.

WHAT YOU WOULD USE

DATA CAPTURE INFRASTRUCTURE

Industrial barcode and RFID (Zebra, Honeywell, Datalogic) — The mature, low-cost capture layer for material and unit identification — the workhorses of manufacturing data capture **(Zebra — vendor; Honeywell — vendor)**.

IoT sensor platforms (PTC ThingWorx, AWS IoT, Litmus) — The aggregation layer that captures PLC, sensor, and equipment state at scale and feeds it to the MES — strongest when the operation has heterogeneous equipment that needs normalization **(PTC — vendor; AWS — vendor)**.

Vision-based capture (Cognex, Keyence, MVTec) — Image-based capture for inspection, defect detection, traceability, and counting — the right answer when the data point is visual and the capture frequency is high **(Cognex — vendor; Keyence — vendor)**.

Direct PLC integration via OPC UA — When the data lives in the PLC (machine state, cycle count, alarm), the cleanest capture is direct PLC tag subscription via OPC UA — no operator involvement, no manual entry, no latency.

4 SO WHAT

> **The operator who is a data entry clerk is not producing — and the data they enter under that condition is not data, it is fiction.** Audit every field, eliminate before instrumenting, match capture technology to data type, design capture into existing motion, validate at the source, and audit data quality continuously. The plants that solve this buy back hours of operator productivity per shift and feed the MES the truth it needs to be useful.

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